

Replacement and Reputation

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Motivation

Primary instrument to control elected officials: **replacement**

Replacement serves two roles:

1. Provide incentives
2. Selection

Other settings: organizational leaders, bureaucrats, etc.

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Our question: Does replacement lead to good outcomes **in the long run**?

This paper

A stylized model of accountability with moral hazard and adverse selection

- Reputational model: long-lived politicians; short-lived voters
- Replacement is the only tool for voters

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there are eqa in which good outcomes do **not** obtain **even in the long run**

Takeaways

- Tension between *guaranteeing* good long-run outcomes and the *possibility* of good outcomes in all periods
- Channels:
 - Guarantee operates through selection
 - Identify source of long-run inefficiencies

Related Literature

Political accountability with selection

- Banks & Sundaram (1993); Fearon (1999); Myerson (2006); Anesi & Buisseret (2022)

Reputation with imperfect monitoring

- Fudenberg & Levine (1992); Cripps, Mailath, Samuelson (2004)

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Reputation with exogenous replacement

- Mailath & Samuelson (2001); Tadelis (2002); Ekmekci, Gossner, Wilson (2012)

Reputation with competition between long-lived players

- Hörner (2002); Atakan & Ekmekci (2015); Deb & Fanning (2024)

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Time: $t = 0, 1, \dots$

Pool of infinitely many identical long-lived politicians

Sequence of short-lived voters, one in each period

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In period 0, one politician is exogenously the **incumbent**

Incumbent chooses **effort** $a_0 \in \{0, 1\}$ and generates public **signal** $s_0 \sim f(\cdot | a_0) \in \Delta S$,

- S is finite, $f(\cdot | 0) \neq f(\cdot | 1)$, and both pmfs have full support

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Period $t \geq 1$:

- Current voter decides **whether to replace the incumbent** with random new draw from pool
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Replacements are public; once a politician is replaced, he never returns

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- Good type exerts effort (action 1) whenever he is in office
- Opportunistic type's stage payoff is

$$u_s = \begin{cases} 0 & \text{if not in office in period } s \\ u(a_s) & \text{if in office in period } s, \end{cases}$$

where $u(0) \equiv 1 > u(1) \equiv 1 - \kappa > 0$. (κ is effort cost)

His overall payoff is $(1 - \delta) \sum_{s=0}^{\infty} \delta^s u_s$, with $\delta \in (0, 1)$

Strategies & Solution Concept

Equilibrium: symmetric weak PBE, i.e., weak PBE in which

- All opportunistic politicians use the same strategy:
sequence of own signals $\rightarrow$ prob of effort
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► why weak PBE

Main Result

(Eventual) First Best

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An equilibrium attains **first best** if $\mathbb{P}(a_t = 1) = 1$ for all $t \geq 0$.

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Note: negative result also under weaker notions of eventual first best

Main Result: An Equivalence Theorem

If an eqm attains FB, it has no learning; so FB must also be attainable absent good type

Intuitively, this requires conjunction of

low-enough effort cost; sufficiently-informative monitoring; enough patience

Formally on next slide → [Condition FB-I](#)

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So, always some eqm that attains Eventual FB

But there is a tension between:

- Eventual FB in all eqa
- FB (or even FB in first period) in some eqm

Condition FB-I

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There is a vector $(v(s))_{s \in S} \in [0, u(0)]^S$ such that

$$(1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f(s|1)v(s) \geq (1 - \delta)1 + \delta \sum_{s \in S} f(s|0)v(s) \quad (\text{IC}_{\text{FB}})$$

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$$(1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f(s|1)v(s) \geq \max_{s \in S} v(s). \quad (\text{PK}_{\text{FB}})$$

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Holds if and only if

- κ is small enough; and
- f is sufficiently informative; and
- δ is large enough

What Prevents Good Long-Run Outcomes?

Theorem says **some** eqa do not attain Eventual FB when FB-I holds

What causes the failure of Eventual FB in those equilibria?

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What causes the failure of Eventual FB in those equilibria?

Let $\pi(h^t)$ denote the incumbent's reputation at history h^t . (Probability of good type.)

Proposition

Consider any equilibrium that does not attain Eventual FB.

There is a positive-prob history h^t at which

$\pi(h^t) > \pi_0$ and yet **the incumbent is replaced with positive prob.**

$\implies$ replacement despite positive track record

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Contrast with the reason for failure of Eventual FB in Myerson (2006)

- There, replacement cost is too high and there is never replacement

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Theorem says **all** eqa attain Eventual FB when Condition FB-I fails.

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Let π_t denote probability that period- t incumbent is a good type.

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In every eqm:

- ① Every opportunistic type is eventually replaced with probability 1.
- ② If FB-I fails, every good type is retained forever with positive probability.

Hence, if FB-I fails, then with probability 1, some good type is never replaced.

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Logic for the first part is closely related to CMS (2004), “impermanent reputations”

Proofs Snippets

Theorem

The following are equivalent:

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I will comment on:

- not (1) $\implies$ not (3)
- (1) $\implies$ (3)

Omit:

- (1) $\iff$ (2) is clear
- (2) [hence (4)] $\implies$ (1)
- (1) $\implies$ (4)

Equilibria that Fail Eventual FB

Why is it not easy to construct eqa that fail Eventual FB?

Naive attempt:

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Solution (when FB-I holds): An eqm in which

- there isn't too much learning
- some shirking in the first period
- in every period the incumbent may be replaced

Hence Eventual FB violated

FB-I Fails $\implies$ Eventual FB in All Equilibria

Two Steps:

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Lemma (Step 1)

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Suppose FB-I fails. In every eqm,
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Lemma (Step 2)

Suppose FB-I fails. In every eqm,
with pos prob the period-0 incumbent is never replaced.

Step 2 $\implies$ a.s., some politician will stay in office forever

$\therefore$ all period-0 incumbents have same prob of never being replaced (symmetric eqm)

Step 1 $\implies$ it is a good type, and hence Eventual FB

Opportunistic Type Eventually Replaced

Lemma (Step 1)

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with probability 1 any opportunistic incumbent is replaced.

Let π_t denote an incumbent's reputation in period t .

- If he is replaced at $s < t$, then let $\pi_t \equiv \pi_s$

Since $\{\pi_t\}_{t \geq 0}$ is a bounded martingale, it converges to some π_∞ a.s.

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Case 1: $\pi_\infty > 0$

- CMS-style logic

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Case 1: $\pi_\infty > 0$

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Case 2: $\pi_\infty = 0$

- because FB-I fails, cannot incentivize effort perpetually, hence eventually replace

Conclusion

Conclusion

A model of accountability with moral hazard and adverse selection

- Each politician is either good or opportunistic
- Replacement is an instrument to provide incentives and to select good politicians

Takeaways

- Replacement makes long-run good outcomes possible (in some eqm) and, sometimes guarantees it (in all eqa)
- Tension between that guarantee and the possibility of good outcomes in all periods
- Source of long-run inefficiencies is replacement despite positive track record

Appendix

Condition FB-I Fails $\implies \mathbb{P}(a_0 = 1) < 1$ in every eqm

If $\mathbb{P}(a_0 = 1) = 1$, then the opportunistic type must be incentivized to play $a_0 = 1$

[\(Back\)](#)

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Consider any $\varepsilon > 0$ small enough.

Because FB-I fails, there exists $s_0 \in S$ such that $V(s_0) \geq V(\emptyset) + \varepsilon$.

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Hence, at history s_0 , the incumbent is retained with pos prob

- Voter optimality $\implies$ the opportunistic type must again be incentivized to play $a_1 = 1$
- There exists $s_1 \in S$ such that $V(s_0, s_1) \geq V(s_0) + \varepsilon$.

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Iterating, for any $t \geq 0$, there is a pos-prob history $(s_0, \dots, s_t)$ s.t. $V(s_0, \dots, s_t) \geq V(\emptyset) + t\varepsilon$

Contradiction, as $V(\cdot) \leq u(0)$

FB-I $\implies$ Equilibria that Attain FB

Consider the following strategy profile

- Incumbent always exerts effort on the eqm path.
- Voter t replaces the incumbent if and only signal s_{t-1} has a LR below some threshold.
- Off path, voter never retains incumbent and incumbent always shirks.

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We show that FB-I $\implies$ there is a threshold s.t. this profile is an eqm.

► Formal

- Only thing to check is incumbent's incentive

Details for FB-I $\implies$ Eqm that Attains FB

Lemma

Assume Condition FB-I. There exist $\bar{v} > 0$ and $x \in (0, 1]$ such that (IC_{FB}) holds and

$$(1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f(s|1)v(s) = \bar{v},$$

with

$$v(s) = \begin{cases} \bar{v} & \text{if } \frac{f(s|1)}{f(s|0)} \geq x \\ 0 & \text{if } \frac{f(s|1)}{f(s|0)} < x. \end{cases}$$

(Back)

Incentive Lemma when FB-I Fails

Lemma

If Condition FB-I fails, then $\exists \varepsilon > 0$ s.t. $\forall (v(s))_{s \in S} \in [0, 1]^S$ that satisfies (IC_{FB}) ,

$$\max_{s \in S} v(s) \geq (1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f(s|1)v(s) + \varepsilon.$$

Strategies & Solution Concept

Equilibrium: *symmetric weak PBE*, i.e., weak PBE in which

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- All opportunistic politicians use the same strategy:
sequence of own signals $\rightarrow$ prob of effort
- All voters use same strategy:
sequence of incumbent's own signals $\rightarrow$ prob of replacement

What are we ruling out?

- Coordination on previous office-holders' signals / when an incumbent took office
- Different politicians using different strategies

$\rightarrow$ *role of replacement is to start interaction with new (ex-ante identical) office-holder afresh*

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Why weak PBE, rather than PBE?

- Strengthens one direction of main result
(property of all equilibria)
- But off-path flexibility plays role in other direction
(construction of a bad equilibrium)

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(construction of a bad equilibrium)

Proposition

A symmetric PBE exists.

Idea: auxiliary game with just one politician; a voter's payoff on replacement is $\pi_0 + (1 - \pi_0)\sigma_P(\emptyset)$

First-Period Outcomes when FB-I Fails

Theorem says that $\mathbb{P}(a_0 = 1) < 1$ in all equilibria when FB-I fails

How bad can period 0 be for voters? (Same as first period for every new incumbent.)

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How bad can period 0 be for voters? (Same as first period for every new incumbent.)

Proposition

Suppose $\{\kappa, \delta, f\}$ violates Condition FB-I.

For every $\varepsilon > 0$, there exists $\bar{\pi}_0 \in (0, 1)$ such that if $\pi_0 < \bar{\pi}_0$, then

$\mathbb{P}(a_0 = 1) < \varepsilon$ in all equilibria.

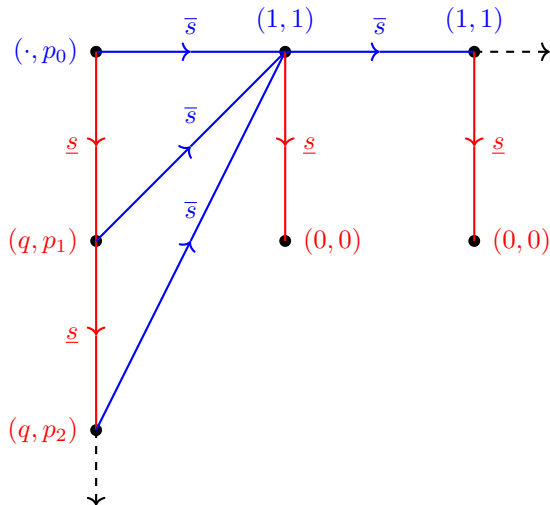
I.e., absent FB-I and if good types are unlikely, then negligible effort from any newly-installed incumbent

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Construction with two signals, $S = \{\underline{s}, \bar{s}\}$.

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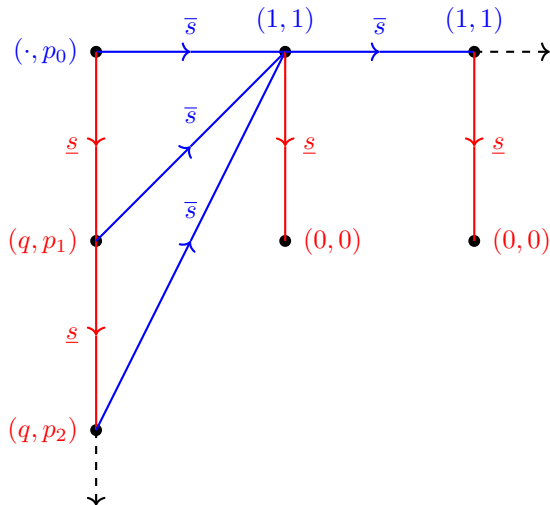
Construction with two signals, $S = \{\underline{s}, \bar{s}\}$. The first number in the vector is the prob of retention, and the second number is the prob of choosing $a = 1$. It holds that $0 < p_0 < p_1 < \dots < 1$ and $q \in [0, 1)$.



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(Back)

Caveat:
use **weak** PBE to sustain
 $(0, 0)$ profiles

Towards Proving Never Replaced With Pos Prob

Let $\mathcal{H}_*$ denote the set of histories s.t.

- The first incumbent reaches that history with positive prob
- The incumbent is retained at that history with positive prob

Let $\bar{S} \equiv \{s \in S | f(s|1) > f(s|0)\}$, i.e., the set of **good signals**.

Lemma

For every $h \in \mathcal{H}_*$, there exists $s \in \bar{S}$ such that $(h, s) \in \mathcal{H}_*$.

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Let $\mathcal{H}_*$ denote the set of histories s.t.

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1. Opportunistic type **does not exert effort at h** .

After observing $s \in \bar{S}$, we have $\pi(h, s) > \pi(h)$.

Since incumbent was retained with pos prob at h , it follows that $(h, s) \in \mathcal{H}_*$.

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Consider two cases:

2. Opportunistic type **exerts effort with pos prob at h** .

Since effort is costly and increases prob of good signals, $V(h, s) > 0$ for some $s \in \bar{S}$, and so $(h, s) \in \mathcal{H}_*$.

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FB-I fails $\implies \exists \varepsilon > 0$ and $s_1 \in S$ such that $V(h, s_1) \geq V(h) + \varepsilon$

Iterating same logic at (h, s_1) and onward, reach a contradiction since $V(\cdot) \leq u(0)$.

Never Replaced With Pos Prob

Lemma (Step 2)

Suppose Condition FB-I fails. In every eqm,
with pos prob the period-0 incumbent is never replaced.

When FB-I fails, $\mathbb{P}(a_0 = 1) < 1$ in all equilibria.

- Another iteration of increasing continuation values argument

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Doob's upcrossing inequality implies that

conditional on $\pi_t \geq 1 - \eta/2$, event $\{\pi_s \geq 1 - \eta \text{ for every } s \geq t\}$ occurs with pos prob.